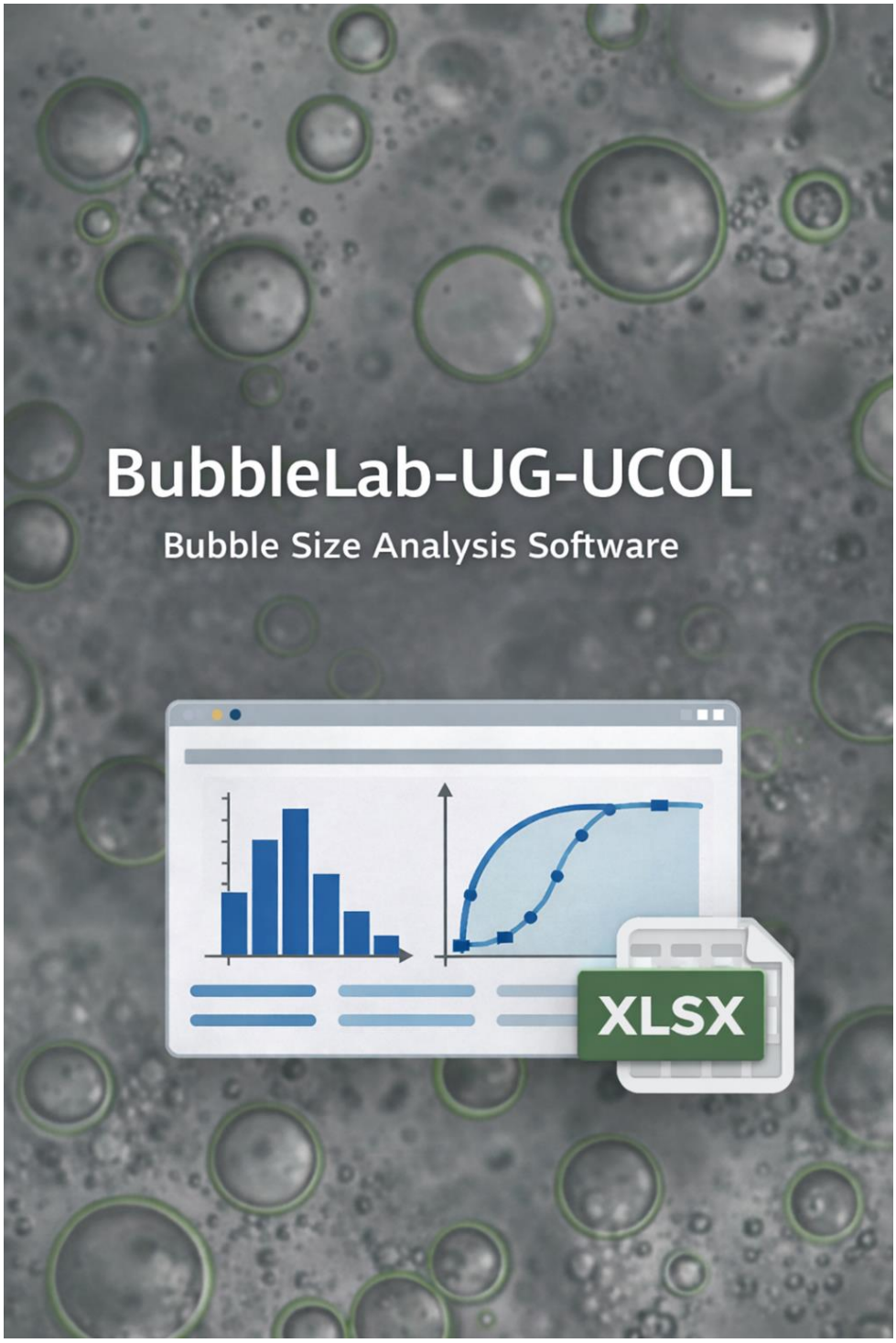


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Software for Bubble Size Distribution Analysis
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# 1. Introduction

BubbleLab-UG-UCOL is an application developed for the automatic analysis of bubble size distributions from digital images obtained in flotation systems or other experiments where bubbles are generated in liquid media.
The software allows the detection of bubbles in individual images or sets of images, measurement of their diameters, and calculation of statistical parameters describing the bubble size distribution present in the system.
Among the calculated parameters are:

- D10
- D50
- D90
- D32 (Sauter mean diameter)

These parameters are widely used in flotation studies because they allow the characterization of the bubble population and the evaluation of operating conditions such as air flow rate, frother type, and hydrodynamic conditions of the system.
BubbleLab-UG-UCOL was developed using Python and computer vision libraries, allowing image processing to be performed rapidly and reproducibly. The application includes tools for:

- spatial scale calibration
- region of interest selection
- automatic batch processing
- visualization of results
- export of data for further analysis

The objective of this software is to provide a simple and reproducible tool for researchers working in mineral processing, chemical engineering, and bubble hydrodynamics studies.
Figure 1 shows the main page of the software, from which the user can upload and process images to obtain the bubble size analysis results.

To access the application, use the following link:

https://bubblelab-ug-ucol.onrender.com/

The software was developed for academic and research purposes in the field of mineral processing.

Figure 1. Main page of the bubble size analysis software (https://bubblelab-ug-ucol.onrender.com/ )

## 2. General Analysis Workflow

The bubble size analysis within the platform is performed through a sequence of six main stages. Each stage corresponds to a specific window in the software and allows the user to progressively prepare, process, and analyze the images.
The stages of the analysis are the following:

Calibration
This stage establishes the relationship between the image scale in pixels and the real physical distance in metric units.

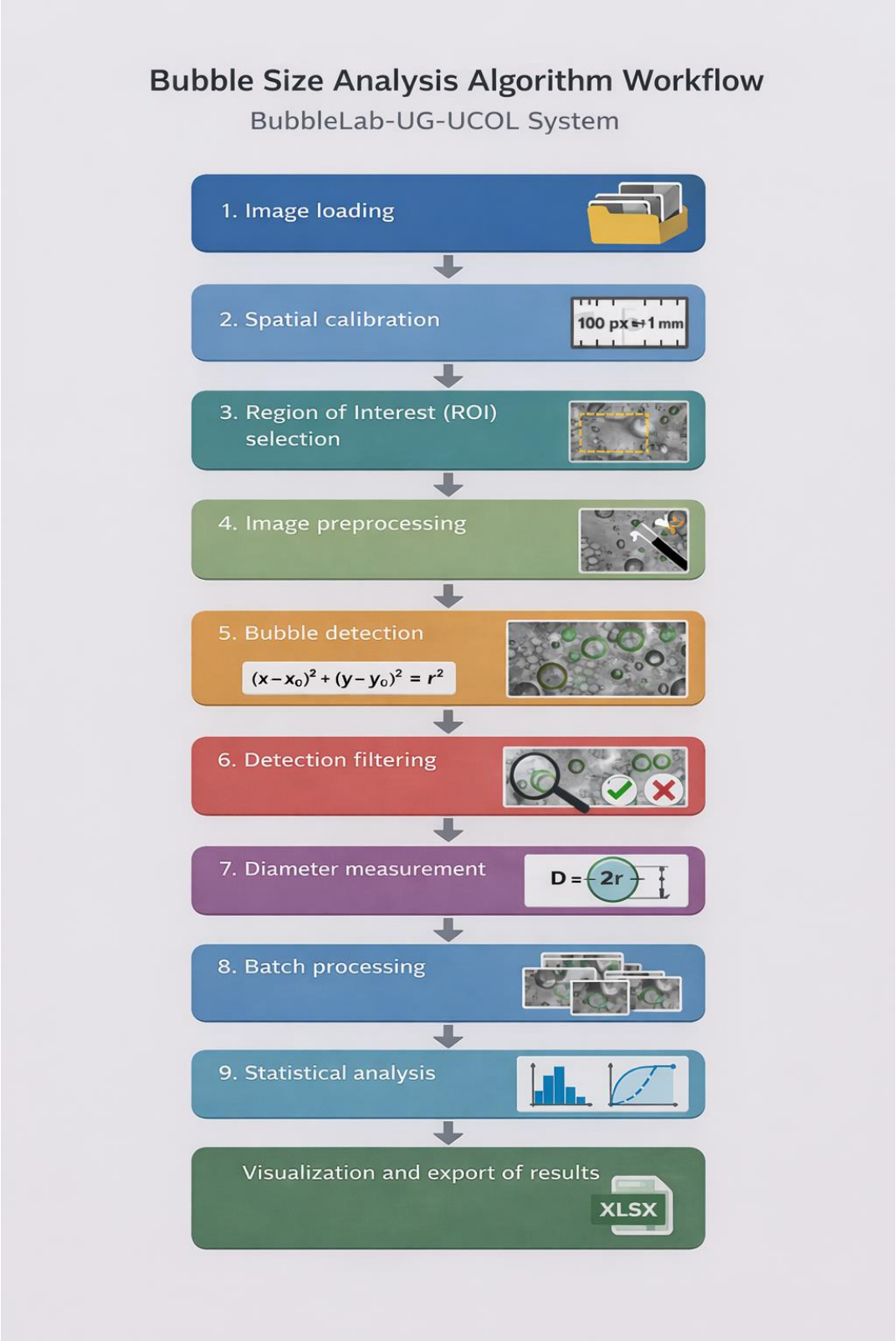
ROI (Region of Interest)
In this stage the user selects the region of the image that will be used for analysis, excluding areas that do not contain bubbles or that may interfere with the image processing.

Adjustments
Image processing parameters are configured in order to correctly identify bubbles within the selected region.

Preview
This stage displays a preliminary visualization of the segmentation results, allowing the user to verify that bubbles have been correctly detected before executing the full analysis.

Batch
Allows the parameters defined previously to be applied automatically to multiple images in order to process complete datasets.

Results
In this section the final analysis results are displayed, including bubble size metrics and the option to export the obtained data.


Bubble Size Analysis Algorithm Workflow
BubbleLab-UG-UCOL System
1. Image loading
2. Spatial calibration
100 px = 1 mm
3. Region of Interest (ROI) selection
4. Image preprocessing
5. Bubble detection
$(x-x_0)^2 + (y-y_0)^2 = r^2$
6. Detection filtering
7. Diameter measurement
D = 2r
8. Batch processing
9. Statistical analysis
Visualization and export of results
XLSX

# 3. Window 1 — Image Upload and Calibration

In this section the user loads the image or set of images to be processed. These images must originate from videos captured using a recording camera capable of registering at least 60 frames per second (fps), which allows proper capture of bubbles in motion within the flotation system.
When extracting images from the video, it is important to ensure that bubbles are not repeated between consecutive frames. For this reason, appropriate frame sampling from the video is recommended, avoiding the selection of images that are too close to each other in time.
It is recommended to organize all images to be analyzed within the same folder and ensure that they were captured under similar illumination and magnification conditions.

Instructions

1. Press Upload Image.
2. Select one or multiple images in JPG or PNG format.
3. The first uploaded image will appear on the screen and will be used to perform the scale calibration, converting pixel units into real metric units.

Recommendations

• Use images with good resolution.
• Ensure that bubbles are clearly visible.
• Avoid strong reflections or saturated illumination areas.

3.1 Scale Calibration

Calibration allows measurements obtained in pixels to be converted into real units (millimeters).

Instructions

1. Draw a line over a known distance within the image.
2. Enter the real distance in millimeters.
3. The software will automatically calculate the conversion factor between pixels and millimeters.

This step is essential to obtain accurate bubble size measurements.
Figure 2 shows an example image that can be used for calibration. In the image a black spherical region corresponding to a steel ball with a known diameter of 1.57 mm is visible and serves as a scale reference.

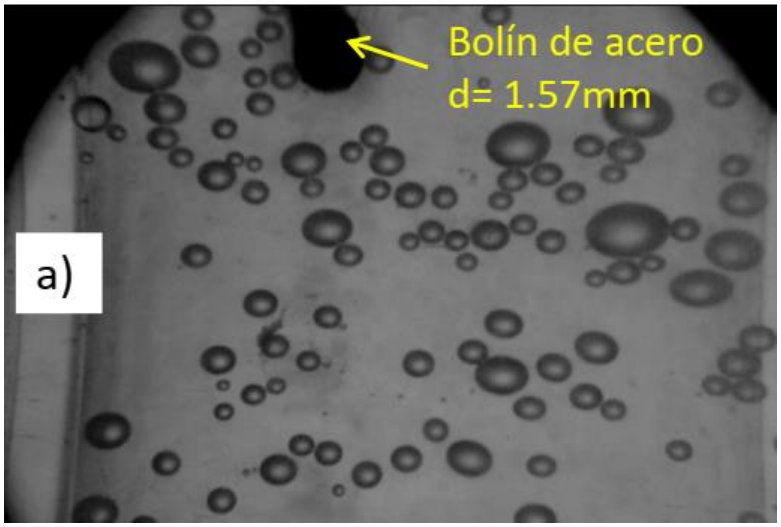



Figure 2. Example image for scale calibration.

Figure 3 shows the red line drawn across the diameter of the steel ball. Once this measurement is completed, a section will appear where the known distance must be entered and the software will automatically perform the scale calibration. This calibration will be applied to all images initially loaded.

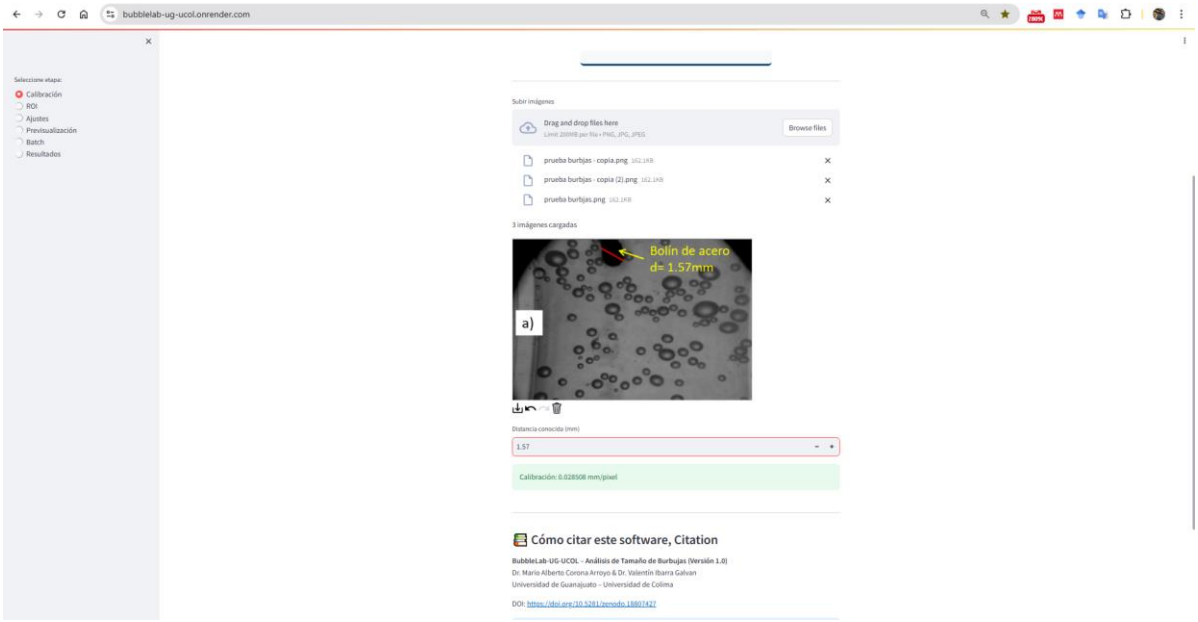



Figure 3. Calibration using a line drawn over a known distance.

## 4. Window 2 — ROI (Region of Interest)

In this section the user can select a Region of Interest (ROI) within the image. This operation allows the user to define the specific area that will be used for the analysis while excluding regions that could introduce interference during the segmentation and image processing stages.

Selecting an appropriate ROI is an important step because it allows the analysis to focus only on the region where bubbles are observed with better focus and stronger contrast relative to the background.

By removing areas with irregular illumination, cell borders, or other undesired elements, the quality of the analysis can be significantly improved.
It is recommended to select a region where:

• Bubbles are clearly defined.
• Good contrast exists between bubbles and background.
• Cell borders, reflections, or out-of-focus areas are avoided.

A proper selection of the ROI facilitates the subsequent processing stages and allows more reliable bubble size measurements to be obtained.

Figure 4 shows a preview of the selected ROI, illustrating the region that will be used for the analysis.

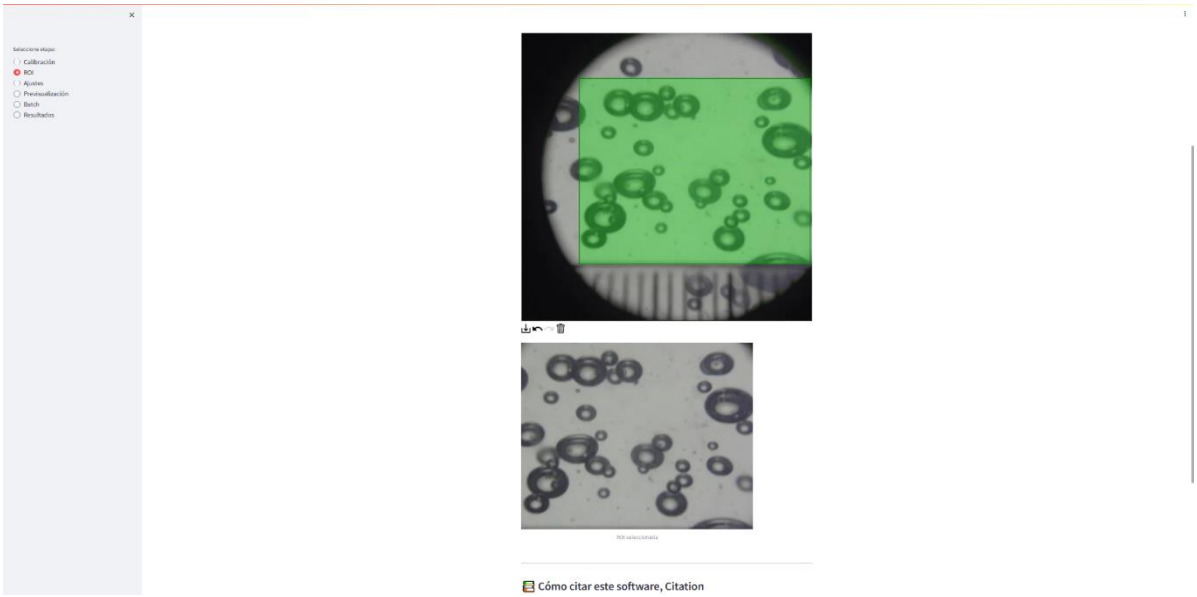



Figure 4. Preview of the selected region of interest (ROI).

## 5. Window 3 — Processing Setting

In this section the user can adjust different image preprocessing and segmentation parameters, which improve the image contrast and facilitate the automatic detection of bubbles.

These parameters directly influence the way the software separates bubbles from the background during the binarization process. Therefore, it is recommended to adjust these controls until a clear and consistent segmentation of bubbles is obtained.

Figure 5 shows the preprocessing adjustment interface, where the sliders corresponding to each parameter are displayed.

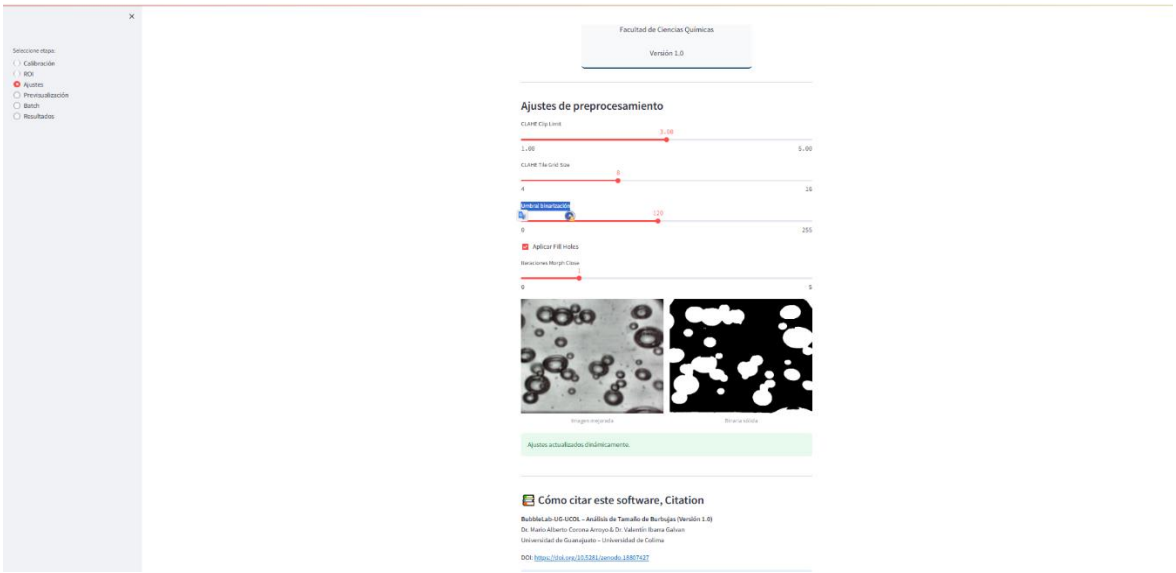



Figure 5. Preprocessing adjustment interface

The adjustments are updated dynamically, allowing the user to observe in real time the effect of each parameter on bubble segmentation.

It is important to consider that the parameters defined in this stage will be applied to all images loaded in the analysis. For this reason, it is recommended that all photographs originate from the same video, ensuring consistent illumination, contrast, magnification, and focus across all images analyzed.

## 5.1 CLAHE Clip Limit

This parameter controls the contrast limit applied during the CLAHE (Contrast Limited Adaptive Histogram Equalization) process.
The CLAHE algorithm improves local contrast in the image, allowing bubble edges to be better distinguished from the background.

- Minimum value: 1.0
- Maximum value: 5.0

Higher values increase image contrast, which may facilitate bubble detection. However, excessively high values may introduce noise or image artifacts.

### 5.2 CLAHE Tile Grid Size

This parameter defines the size of the regions (tiles) into which the image is divided when applying the CLAHE algorithm.

- Minimum value: 4
- Maximum value: 16

A smaller value means that contrast is adjusted in smaller regions, improving local details. Larger values produce smoother contrast adjustment across the image.

### 5.3 Binarization Threshold

The binarization threshold defines the intensity value used to convert the image into a binary image (black and white).

Pixels above the threshold become white (bubble regions). Pixels below the threshold become black (background).

- Minimum value: 0
- Maximum value: 255

A threshold that is too low may generate false detections, while a threshold that is too high may prevent some bubbles from being fully detected.

### 5.4 Fill Holes

This option applies a hole-filling operation within the detected bubble regions.

During segmentation, bubbles may appear with internal dark regions due to illumination variations. The Fill Holes option fills these internal regions, producing solid bubbles in the binary image.

This improves the measurement of bubble area and equivalent diameter.

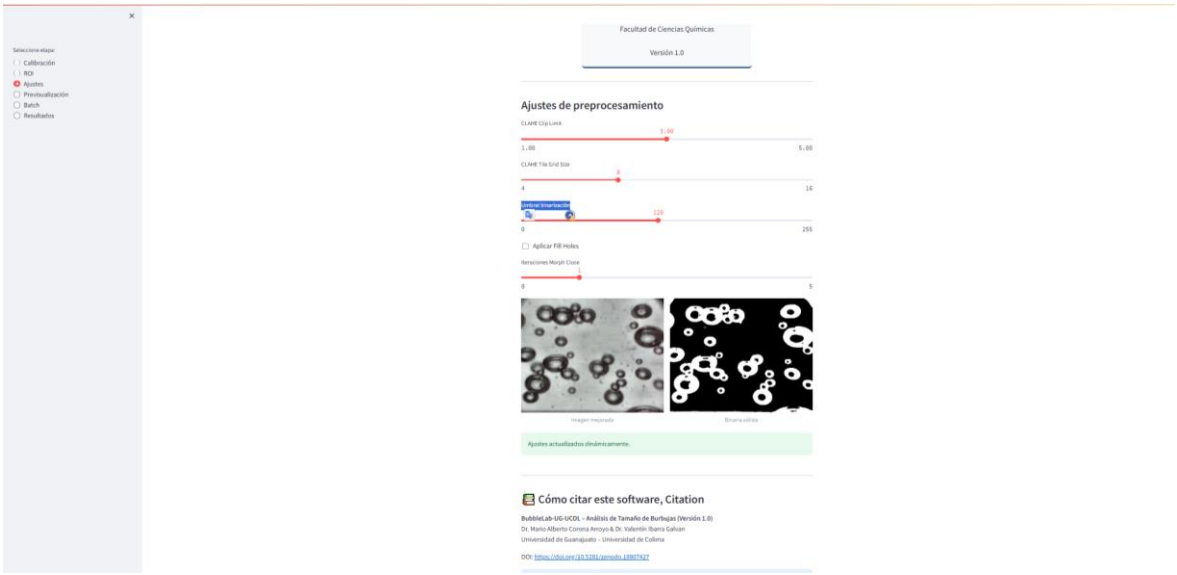



Figure 6. Example of the effect of the Fill Holes filter on the binary image. When this filter is not applied, the bubbles show a hole in the center due to the illumination effect produced by the position of the lighting lamp.

### 5.5 Iteraciones Morph Close

This parameter controls the number of iterations of the morphological closing operation applied to the binary image.
Morphological closing combines:

- Dilation
- erosion

Its objective is to:

- close small holes
- connect nearby regions
- smooth bubble contours
- Minimum value: 0
- Maximum value: 5

However, excessively high values may cause nearby bubbles to merge incorrectly.

## 6. Ventana 4 — Previsualización

In this section the software allows the user to visually evaluate the automatic bubble detection before executing the full analysis. The tool applies the bubble identification algorithm to the segmented image and displays green circles representing the detected bubbles.

Figure 7 shows the interface of the preview section, where the filtering parameters, bubble separation control, and the “Preview segmentation” button can be observed. This button allows the user to run the detection algorithm.

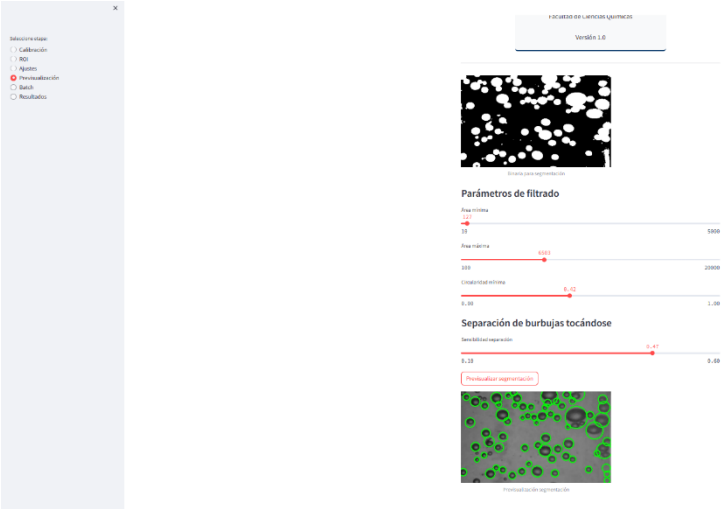



Figure 7. Interface of the preview and bubble detection section.

From the beginning it is recommended to press the “Preview segmentation” button, as this will display the green circles identifying each bubble detected by the algorithm. This visualization allows the user to quickly evaluate whether the segmentation and selected parameters are correctly identifying the bubbles present in the image.

Each time any parameter in this section is modified, it is important to press the preview button again. This will update the detection results and allow the user to observe how the identification of bubbles changes.

The objective of this stage is to optimize the parameters of the detection algorithm, ensuring that each bubble is correctly identified as an individual entity before executing the full processing of the image dataset

## 6.1 Filtering Parameters

*Minimum Area*

This parameter defines the minimum size of detected regions that will be considered valid bubbles.

- Minimum value: 10
- Maximum value: 5000

Regions smaller than the defined value will be automatically discarded. This helps eliminate noise or small regions incorrectly detected during segmentation.
Values that are too low may allow detection of artifacts or noise, while values that are too high may eliminate real small bubbles.

*Maximum Area*
This parameter defines the maximum allowed size for a region to be identified as a bubble.

- Minimum value: 100
- Maximum value: 20000

This filter removes regions that are excessively large, which usually correspond to grouped bubbles, viewer borders, or segmentation errors.

Adjusting this parameter improves the quality of the bubble dataset used for analysis.

*Maximum Circularity*
This parameter controls the degree of circularity accepted for regions detected as bubbles.
Circularity is a geometric measure that evaluates how close the shape of an object is to a perfect circle.

- Minimum value: 0.00
- Maximum value: 1.00

A value close to 1 corresponds to highly circular shapes, while lower values allow more irregular shapes to be included.

This parameter helps eliminate regions that do not correspond to real bubbles, since bubbles observed in flotation systems typically present approximately circular shapes.

*Separation of Touching Bubbles-Separation Sensitivity*

This parameter controls the sensitivity of the algorithm used to separate bubbles that are in contact or very close to each other.

- Minimum value: 0.10
- Maximum value: 0.60

Higher values increase the algorithm's ability to detect and separate bubbles that are partially overlapping or touching each other. However, excessively high values may lead to over-segmentation, where a single bubble is incorrectly divided into multiple regions.

This parameter is particularly important in flotation systems where bubbles tend to interact and form clusters.

## 6.2 Funcionamiento del algoritmo de detección de burbujas

The algorithm implemented in the software combines several image processing stages to achieve robust bubble detection. The main stages include:

- Contrast enhancement using CLAHE
- Image binarization to separate bubbles from the background
- Application of morphological operations to clean the segmentation
- Hole filling (Fill Holes) to obtain solid bubbles
- Geometric filtering based on area and circularity
- Separation of touching bubbles using segmentation techniques

Once this preprocessing is completed, the software applies a circle detection procedure based on contour analysis of the segmented regions.

This process uses widely adopted Python image processing libraries such as OpenCV and Scikit-Image, which allow the identification of connected regions within the binary image and the calculation of their geometric properties.

Among the analyzed properties are:

- Region area
- Perimeter
- Centroid
- Equivalent radius
- Circularity

Bubble identification is performed using the Circular Hough Transform, a widely used method for detecting circular structures in images.

This method identifies circles in the image by evaluating the pixels that satisfy the geometric equation of a circle:

$$(x - x_c)^2 + (y - y_c)^2 = r^2$$

where:

$x_c, y_c$represent the center of the circle
$r$represents the radius of the bubble

The algorithm analyzes the image and determines the locations where there is a high probability of circular structures corresponding to bubbles.

As a result of this stage, the following information is obtained:

- position of the center of each bubble
- radius of each detected bubble

From these properties, the algorithm determines whether a region corresponds to a valid bubble. Subsequently, an equivalent circle is calculated to geometrically represent each bubble, which is displayed in the image using the green identification circles.

The diameter of each bubble is obtained from the equivalent diameter of a circle with the same area as the segmented region, allowing a consistent bubble size measurement even when the bubble contour is not perfectly circular.

The combination of all these stages makes the algorithm robust against variations in illumination, contrast, and minor segmentation imperfections, allowing reliable bubble size measurements.

The preview stage allows the user to visually validate this process before executing the full analysis on all loaded images.

## 7. Window 5 — Batch Processing

In this section the software performs the automatic processing of all loaded images, using the segmentation and detection parameters defined in the previous stages.

The objective of this stage is to apply the bubble detection algorithm consistently across the entire image dataset, allowing the generation of a statistically representative dataset of bubble sizes.
Figure 8 shows the interface of the Batch section, where the preliminary results of the image processing are displayed.

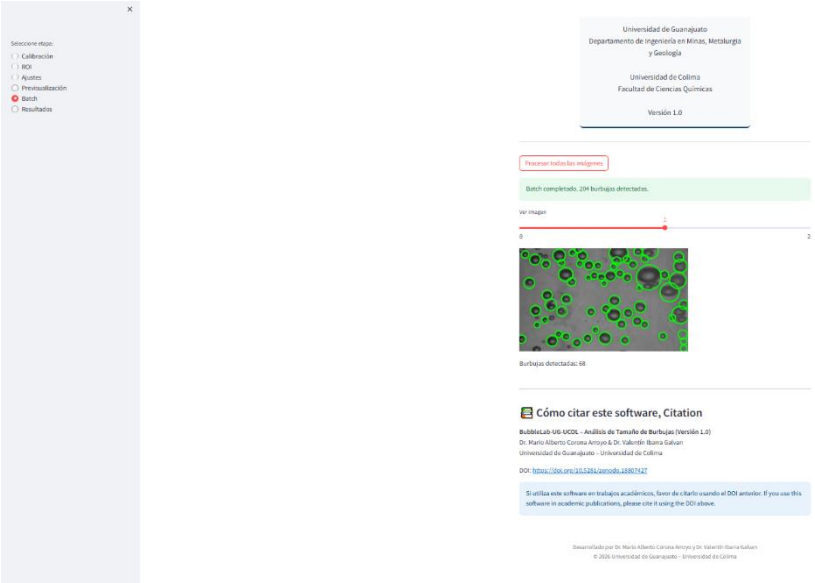



Figure 8. Interface of the Batch section for reviewing the image processing results.

*Information displayed in this section*
*Number of bubbles detected per image*

The software displays the number of bubbles detected in each processed image. This information allows the user to verify that the algorithm is correctly identifying the bubbles in every image of the dataset.

*Preview of the analysis per image*

This section also allows the user to preview how the measurements were performed for each image. Using the scroll bar (slider), the user can browse through all processed images and observe the detection results for each one.

In each image, green circles indicate the bubbles identified by the algorithm, allowing a visual verification of the detection quality.

This functionality is particularly useful for confirming that the selected parameters work correctly for the entire image dataset, and not only for the image used during the preview stage.

Total number of bubbles detected

In addition to the count per image, the software also displays the total number of bubbles detected across all processed images.

This value represents the total size of the bubble population analyzed, and it will later be used to calculate bubble size statistics and the diameter distribution.

*Global verification of the analysis*

The Batch section allows a global review of the bubble identification process, ensuring that the algorithm has performed correctly across all images in the dataset.
It is recommended to review several images using the slider to confirm that:

- Bubbles are correctly identified
- There are no significant false detections.
- Touching bubbles have been properly separated.

Once the detection quality has been verified across the image dataset, the user can proceed to the final Results stage, where the statistical metrics of bubble size are calculated.

## 8. Window 6 — Analysis Results

In this section the software presents the final results of the bubble size analysis, obtained from the processing of all images loaded in the analysis.

The results are calculated using the complete set of bubbles detected during the Batch processing stage. Based on this information, the software generates statistical parameters and graphical representations that allow interpretation of the bubble size distribution of the analyzed system.

Figure 9 shows the interface of the Results section, where the main statistics, bubble size distribution plots, and the option to export the data are displayed.

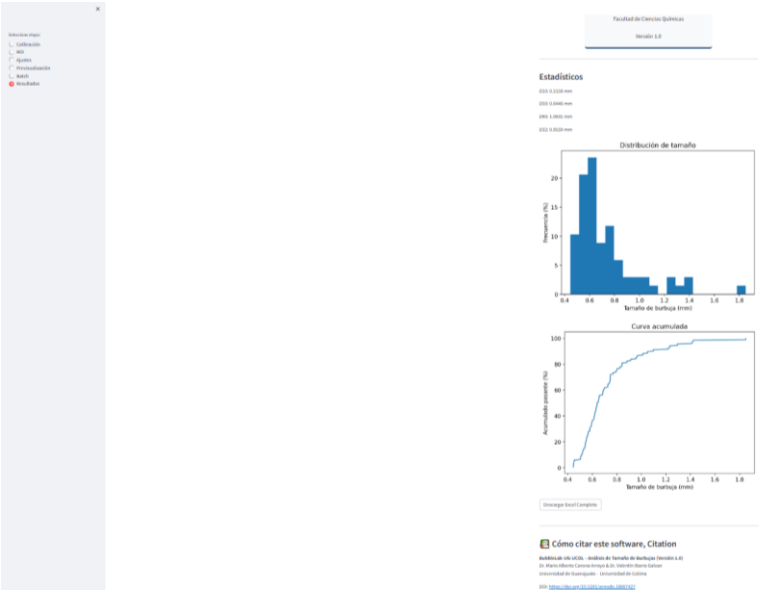



Figure 9. Interface of the results section of the bubble size analysis.

*Statistical parameters*

The software calculates several statistical parameters commonly used in the analysis of bubble size distributions.

- D10. The D10 parameter represents the diameter for which 10% of the bubbles have a size equal to or smaller than this value. This parameter characterizes the fraction of smaller bubbles within the distribution.
- D50. The D50 parameter, also known as the median of the distribution, corresponds to the diameter for which 50% of the bubbles have a size equal to or smaller than this value. This value represents the characteristic size of the bubble population and is one of the most commonly used indicators to describe the distribution.
- D90.The D90 parameter represents the diameter for which 90% of the bubbles have a size equal to or smaller than this value. This parameter characterizes the presence of larger bubbles within the distribution.

- D32 — Sauter Mean Diameter. The D32 parameter, known as the Sauter mean diameter, represents the diameter of a bubble that has the same volume-to-surface ratio as the entire bubble population.

This parameter is widely used in flotation studies and interfacial phenomena because it is directly related to the surface area available for interaction between bubbles and particles.
The Sauter diameter is calculated using the following relationship:

$$D_{32} = \frac{\sum d_i^3}{\sum d_i^2}$$

where $d_i$corresponds to the equivalent diameter of each detected bubble.

*Graphical representations*

The software generates two main graphs that allow visualization of the bubble size distribution.

*Bubble size distribution histogram*

The first graph corresponds to a bubble size distribution histogram, where the frequency of bubbles within different diameter intervals is displayed.

This graph allows the user to observe:

- the range of bubble sizes present
- the predominant bubble size
- the overall shape of the distribution

*Cumulative distribution curve*

The second graph shows the cumulative bubble size distribution curve. This curve represents the cumulative percentage of bubbles as a function of their diameter, which allows easy determination of the D10, D50, and D90 values.
Exporting results
At the bottom of this section there is a “Download results” button that allows exporting the analysis data in Excel format (.xlsx).

The exported file includes:

- Equivalent diameter of each bubble
- Area of each bubble
- Distribution statistics
- Total number of analyzed bubbles

This functionality allows additional analysis or complementary statistical processing using other data analysis tools.

## 9. About the Author

Dr. Mario Alberto Corona Arroyo is a professor and researcher in the Department of Mining, Metallurgical and Geological Engineering at the University of Guanajuato, Mexico, where he conducts research in the field of mineral processing and extractive metallurgy.

His scientific work focuses on the study of physicochemical phenomena in flotation processes, mineral surface characterization, and the development of experimental methods to improve the efficiency of mineral separation processes. His research interests include flotation chemistry, mechanisms of reagent action, magnetite beneficiation, and combined flotation and magnetic separation processes.

Dr. Corona Arroyo has authored several scientific publications in the areas of mineral processing, surface chemistry, and mineral characterization, and participates in research projects aimed at developing technologies for the mining and metallurgical industries.
He currently serves as a member of the Early Career Editorial Board of the international journal *Minerals Engineering*, one of the leading scientific journals in the field of mineral processing and extractive metallurgy.

The development of the BubbleLab-UG-UCOL software arose from the need to provide accessible tools for the automatic analysis of bubble size distribution in flotation systems, using computer vision techniques and digital image processing. This software was designed for academic and research purposes, with the objective of facilitating the reproducible analysis of bubble images obtained from laboratory experiments.

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